THE ASPECT RATIO OF THE TWIN DRAGON IS $1/\varphi$

DMITRY MEKHONTSEV

Abstract. We show that the geometric aspect ratio of the Twin Dragon equals $1/\varphi$, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio. The result follows by solving the covariance fixed-point equation for the self-similar measure, which coincides with Lebesgue area since the similarity dimension is 2. The appearance of φ is surprising: the Twin Dragon is defined purely via the Gaussian integer $1+i$, with no pentagonal or Fibonacci structure in its construction.

1. Introduction

The Twin Dragon is a classical plane-filling fractal [4] governed by the Gaussian integer $1+i$: its two defining contractions share the linear part $(1-i)/2$, and their translations are symmetric about the origin. Despite this simple, purely arithmetic construction, the attractor has a non-trivial shape that is not immediately obvious from the definition.

A natural quantitative measure of shape for a planar set is the *geometric aspect ratio*: the ratio of standard deviations along the principal axes of its area distribution, or equivalently the square root of the ratio of the eigenvalues of the second-moment (covariance) matrix M of the uniform measure on the attractor. This definition gives the expected answers for elementary shapes: a circle has aspect ratio 1 (all eigenvalues equal); an $a \times b$ rectangle has aspect ratio a/b (eigenvalues $a^2/12$ and $b^2/12$); and a square has aspect ratio 1.

The approach of characterising IFS attractors through moments of their invariant measures was introduced by Vrscay and Roehrig [6] and systematically developed in [7]. The key observation is that the self-similar measure satisfies a linear fixed-point equation in its moments, which can be solved exactly. In favourable cases — when the similarity dimension equals the ambient dimension — this gives the moments of the actual area measure on the attractor.

We apply this method to the Twin Dragon. The covariance matrix turns out to be $M = \frac{1}{5}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$, whose eigenvalues involve $\sqrt{5}$, leading to the aspect ratio $1/\varphi$, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio. The appearance of φ in a fractal with no pentagonal geometry is unexpected.

Discovery. The result was first found empirically using the IFStile software [5], which implements the moment invariants of [6, 7] as part of its algebraic IFS search engine. Among the similarity invariants reported by IFStile for the Twin Dragon,

Date: April 6, 2026.

2020 *Mathematics Subject Classification.* 28A80, 37C45, 15A18.

Key words and phrases. Twin Dragon, iterated function system, self-similar measure, second moments, golden ratio, aspect ratio.

the aspect ratio $1/\varphi$ appeared as a numerical value; the present paper supplies the exact derivation.

The paper is organised as follows. Section 2 recalls the definition of the Twin Dragon and its IFS. Section 3 derives the covariance matrix by solving the fixed-point equation. Section 4 computes the eigenvalues and states the main result.

2. The IFS

An iterated function system (IFS) [2] is a finite collection of contractions on a complete metric space; by Hutchinson's theorem [2] there exists a unique non-empty compact attractor. For background on fractal geometry and self-similar sets see [3].

The Twin Dragon $A \subset \mathbb{R}^2$ is the attractor of the IFS [4]

$$A = f_1(A) \cup f_2(A), \qquad f_{1,2}(z) = \frac{1-i}{2}(z \mp 1), \tag{1}$$

where $z = x + iy$. In matrix form, both maps share the same linear part

$$A_0 = g^{-1} = \tfrac{1}{2}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \qquad g = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \tag{2}$$

with translation vectors $b_1 = (-\frac{1}{2}, \frac{1}{2})^T$ and $b_2 = -b_1$. The contraction ratio is $\|A_0\| = 1/\sqrt{2}$, and the similarity dimension is $\alpha = 2$ (positive area).

3. Covariance of the Invariant Measure

Vrscay and Roehrig [6] observed that the moments of the invariant measure of an IFS satisfy linear recursion relations that can be solved exactly. For a self-similar IFS $\{f_i\}$ with invariant measure μ, the covariance matrix $M = \int xx^T \, d\mu$ satisfies a fixed-point equation linear in M, which is uniquely solvable whenever the spectral radius of the associated linear operator is less than one (see [7] for a systematic treatment).

For the Twin Dragon, the situation is particularly clean. The open set condition holds [2], so by [5, §10.3] (see also [2, Theorem 5.3]) the self-similar measure μ is proportional to $\mathcal{H}^2|_A$. Since the similarity dimension equals the ambient dimension $\alpha = 2$, the Hausdorff 2-measure coincides with Lebesgue area up to a constant, so $\mu \propto \mathcal{L}^2|_A$: the covariance M describes the actual geometric shape of the attractor.

The measure μ is the unique probability measure satisfying $\mu = \frac{1}{2}(f_1)_*\mu + \frac{1}{2}(f_2)_*\mu$ [2] (equal weights because both maps have contraction ratio $r = 1/\sqrt{2}$, hence $r^\alpha = \frac{1}{2}$). Since $b_1 + b_2 = 0$, the centre of mass is $c = 0$, and the general recursion specialises to

$$M = A_0 \, M \, A_0^T + b_1 b_1^T. \tag{3}$$

This form of the Euler tensor equation for self-similar IFS is given in [1, Theorem 3.3] and applied in [5, §11.3].

Translation term.

$$b_1 b_1^T = \frac{1}{4}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \tag{4}$$

Solving the linear system. Write $M = \begin{pmatrix} p & q \\ q & r \end{pmatrix}$. Expanding $A_0 M A_0^T$:

$$A_0 M A_0^T = \frac{1}{4}\begin{pmatrix} p+2q+r & -p+r \\ -p+r & p-2q+r \end{pmatrix}. \tag{5}$$

Substituting into (3) and collecting terms:

$$3p - 2q - r = 1, \tag{6}$$

$$p + 4q - r = -1, \tag{7}$$

$$p - 2q - 3r = -1. \tag{8}$$

The unique solution is $p = \frac{2}{5}$, $q = -\frac{1}{5}$, $r = \frac{3}{5}$, giving

$$M = \frac{1}{5}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}. \tag{9}$$

4. Eigenvalues and Aspect Ratio

The matrix M has $\operatorname{tr} M = 1$ and $\det M = \frac{1}{5}$, so its eigenvalues are

$$I_{1,2} = \frac{1 \mp \frac{1}{\sqrt{5}}}{2}. \tag{10}$$

The geometric aspect ratio — the ratio of standard deviations along the principal axes of M [5, Proposition 45] — is

$$\sqrt{\frac{I_1}{I_2}} = \sqrt{\frac{\sqrt{5}-1}{\sqrt{5}+1}} = \sqrt{\frac{3-\sqrt{5}}{2}} = \frac{1}{\varphi}, \tag{11}$$

where $\varphi = \frac{1+\sqrt{5}}{2}$ is the golden ratio.

Theorem 1. *The geometric aspect ratio of the Twin Dragon — the ratio of standard deviations along the principal axes of its covariance matrix M — equals $1/\varphi$, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio.*

Remark 1. *The Twin Dragon is governed by the Gaussian integer $1+i$: its two maps are simply $f_{1,2}(z) = \frac{1-i}{2}(z \mp 1)$, with no pentagon, regular decagon, or Fibonacci recurrence in the construction. The emergence of φ is therefore unexpected.*

References

[1] A. DiCarlo and A. Paoluzzi, *Fast computation of inertia through affinely extended Euler tensor*, Comput.-Aided Des. **38** (2006), no. 11, 1145–1153. doi:10.1016/j.cad.2006.09.003.

[2] J. E. Hutchinson, *Fractals and self-similarity*, Indiana Univ. Math. J. **30** (1981), no. 5, 713–747. doi:10.1512/iumj.1981.30.30055.

[3] K. J. Falconer, *Fractal Geometry: Mathematical Foundations and Applications*, 3rd ed., Wiley, 2014.

[4] C. Davis and D. E. Knuth, *Number representations and dragon curves*, Journal of Recreational Mathematics **3** (1970), no. 2–4, 66–81, 133–149.

[5] D. Mekhontsev, *An algebraic framework for finding and analyzing self-affine tiles and fractals*, Doctoral thesis, Ernst-Moritz-Arndt-Universität Greifswald, 2019. urn:nbn:de:gbv:9-opus-24794.

[6] E. R. Vrscay and C. J. Roehrig, *Iterated function systems and the inverse problem of fractal construction using moments*, in *Computers and Mathematics* (E. Kaltofen and S. M. Watt, eds.), Springer, New York, 1989, pp. 250–259. doi:10.1007/978-1-4613-9647-5_29.

[7] E. R. Vrscay, *Iterated function systems: theory, applications and the inverse problem*, in *Fractal Geometry and Analysis* (J. Bélair and S. Dubuc, eds.), Kluwer, Dordrecht, 1991, pp. 405–468. doi:10.1007/978-94-015-7931-5_10.

Novosibirsk State University, Novosibirsk, Russia
Email address: mekhontsev@gmail.com